# STANDATA interpretation 23-BCI-026/23-FCI-014

Building/Fire

## Single-Egress-Stairs (SES) for buildings of residential occupancy

Date Issued: October 2026



### Purpose

To clarify that compliance with the National Building Code – 2023 Alberta Edition (NBC(AE)) can be achieved via alternative solutions (variance) that will achieve at least the minimum level of performance required by Division B in the areas defined by the objectives and functional statements attributed to the applicable acceptable solutions.

Where a single-egress-stair (SES) building does not comply with the acceptable solutions of Division B, this STANDATA clarifies that it is possible for the building to comply with the NBC(AE) via alternative solutions.

### Discussion

With growing demands to densify urban areas to address affordable housing, designers are exploring different methods to satisfy the requests of developers and building owners. One such design that enables smaller residential lots in built-up urban areas to be fully utilized is the SES building. This style of building challenges the conventional multi-unit residential building design (comprised of two or more egress stairs), while still supporting multiple dwelling units in a building.

Commonly referred to a “SES” building, some designers identified such buildings as a potential answer to the “missing middle” (a concept that highlights a lack of medium-density housing options). Alberta cities need medium-density housing throughout their urban areas, while these urban areas also contain vacant lots not large enough to construct conventional, high-density residential buildings. Designers consequently propose this as an opportunity to solve an urban planning and housing problem via smaller-footprint SES buildings.

Change is being sought to introduce prescriptive requirements for SES buildings into a future edition of the National Building Code of Canada. Until such changes are established, designers continue with the current convention of two or more egress stairs for buildings three or more storeys in building height or propose an SES building design in the form of an alternative solution.

Alternative solutions are a valid method of demonstrating compliance with the NBC(AE), as per Sentence 1.2.1.1.(1) of Division A. Permit applications which include an Alternative Solution for SES buildings must be accepted by authorities having jurisdiction (AHJ) in accordance with the *Safety Codes Act* (Act).

**AHJ Issuance or Refusal of a Variance for an Alternative Solution Application**

A building safety codes officer (building SCO) may refuse to issue a variance for an alternative solution application for the following reasons, which include, but are not limited to:

- There is insufficient or incomplete information,
- The input parameters or assumptions are incorrect,
- The intent of an alternative solution application is not demonstrated in the analysis (approximately equivalent or greater safety performance), or



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- The recommendation is unclear, or impractical/unachievable (i.e. constraints are impossible to achieve or affect other tenancies or buildings).

If a building SCO refuses to issue a variance, the applicant must be served with a written notice of the refusal. The applicant may appeal against the refusal of the variance to the Safety Codes Council (Council) within 30 days after the date on which the notice was received (see sections 38(6) and 38(7) of the Act).

**Professional Involvement**

Whether professional involvement is required for a building's design depends on its features, such as, but not limited to, occupancy type, number of storeys, number of dwelling units, and building area. Where compliance with the NBC(AE) is being addressed via an alternative solution submission, the qualification of the person developing the alternative solution may include a registered professional but is not prescribed as such. Subclause 2.4.1.1.(1)(b)(ii) of Division C requires professional involvement for buildings containing more than 4 dwelling units.

**National Fire Code – 2023 Alberta Edition**

After occupancy is granted for an SES building designed via the alternative solution compliance route, documentation conforming to Article 2.3.1.1. of Division C of the National Fire Code – 2023 Alberta Edition (NFC(AE)) must be retained on the premises for review by fire safety codes officers (fire SCOs). This is required by Clause 2.3.1.2.(1)(b) of the NFC(AE) and serves as a manner of verifying that the conditions under which the variance was issued have not been contravened.

## Safety Codes Act References

***Safety Codes Act* Part 2 Administration**

**Inspections**

34(1) For the purpose of ensuring that this Act and any thing issued under this Act are complied with, a safety codes officer may, without a warrant, at any reasonable time, enter any premises or place, except a private dwelling place that is in use as a dwelling, in which the officer has reason to believe there is something to which this Act applies and may, using reasonable care, carry out an inspection, review designs and examine and evaluate quality management systems and manufacturing and construction processes.

***Safety Codes Act* Part 2 Administration**

**Production of documents**

35(1) For the purpose of ensuring that this Act and anything issued under this Act are complied with, a safety codes officer may demand the production, within a reasonable time, of any record or document pertaining in any manner to compliance with this Act and may on giving a receipt for it remove it for not more than 48 hours for the purpose of making copies of it.



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***Safety Codes Act* Part 3 Standards**

**Variances**

38(1) An Administrator or a safety codes officer may issue a written variance with respect to any thing, process or activity to which this Act applies if the Administrator or officer is of the opinion that the variance provides approximately equivalent or greater safety performance with respect to persons and property as that provided for by this Act.

(2) An Administrator or a safety codes officer may include terms and conditions in the variance.

(3) A safety codes officer on issuing a variance shall notify an Administrator.

(4) The *Regulations Act* does not apply to variances issued under this section.

(5) A decision of the Administrator under subsection (1) is final and not subject to appeal.

(6) If a safety codes officer refuses to issue a written variance, the applicant shall be served with a written notice of the refusal.

(7) An applicant who receives a notice referred to in subsection (6) may, within 30 days after the date on which the notice is received, appeal the refusal to the Council in accordance with the Council's bylaws.

***Safety Codes Act* Part 5 Orders, Appeals**

**Appeal of refusals, suspensions, cancellations**

51(1) The Council, on receipt of a notice of appeal with respect to

(a) a refusal to designate a corporation as an accredited corporation or a person as an accredited agency,

(a.1) a refusal by a safety codes officer to issue a written variance,

(b) a refusal to register a design or a deregistration of a design, or

(c) a suspension or cancellation of a designation of accreditation, a certificate of competency or a permit,

shall send a copy of the notice of appeal to the relevant Administrator and the safety codes officer, if any, who issued the suspension or cancellation, and notify them and the appellant of the time and place of the appeal.

(2) In order for an appeal to proceed, the Council must receive a notice of appeal in the form approved by the Council within 30 days after the date the corporation or person was served with the written notice of the refusal to designate, refusal by a safety codes officer to issue a written variance, refusal to register, deregistration, suspension or cancellation.



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## Code References

### NBC(AE) Preface

In Alberta, the Safety Codes Act recognizes “alternative solutions” as Variances. The Safety Codes Act has provisions to ensure that Variances are issued by appropriate individuals and registered with the Safety Codes Council. The Safety Codes Council can provide additional information on policy and procedures regarding the issuance of Variances.

### NBC(AE) Division A, Article 1.2.1.1.

**1.2.1.1. Compliance with this Code**

1) Compliance with this Code shall be achieved by
   a) complying with the applicable acceptable solutions in Division B (see Note A-1.2.1.1.(1)(a)), or
   b) using alternative solutions that will achieve at least the minimum level of performance required by Division B in the areas defined by the objectives and functional statements attributed to the applicable acceptable solutions (see Note A-1.2.1.1.(1)(b)).

### NBC(AE) Division A, Article 1.4.1.2.

**1.4.1.2. Defined Terms**

*Authority having jurisdiction* means a safety codes officer in the building discipline exercising authority pursuant to designation of powers and terms of employment in accordance with the Safety Codes Act.

*Building* means any structure used or intended for supporting or sheltering any use or *occupancy*.

*Building area* means the greatest horizontal area of a *building* above *grade* within the outside surface of exterior walls or within the outside surface of exterior walls and the centre line of *firewalls*.

*Building height* (in *storeys*) means the number of *storeys* contained between the roof and the floor of the *first storey*.

*Exit* means that part of a *means of egress*, including doorways, that leads from the *floor area* it serves to a separate *building*, an open public thoroughfare, or an exterior open space protected from fire exposure from the *building* and having access to an open public thoroughfare.

*Floor area* means the space on any *storey* of a *building* between exterior walls and required *firewalls*, including the space occupied by interior walls and *partitions*, but not including *exits*, *vertical service spaces*, and their enclosing assemblies.

*Means of egress* means a continuous path of travel provided for the escape of persons from any point in a *building* or contained open space to a separate *building*, an open public thoroughfare, or an exterior open space protected from fire exposure from the *building* and having access to an open public thoroughfare. *Means of egress* includes *exits* and *access to exits*.

*Registered architectural professional* means an individual who is authorized to engage in the practice of architecture under the Architects Act and its Regulations.

*Registered engineering professional* means an individual who is authorized to engage in the practice of engineering under the Engineering and Geoscience Professions Act and its Regulations.



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*Registered professional* means an individual who qualifies as a
(a) *registered architectural professional*,
(b) *registered engineering professional*, or
(c) *licensed interior designer*.

*Storey* means that portion of a *building* that is situated between the top of any floor and the top of the floor next above it, and if there is no floor above it, that portion between the top of such floor and the ceiling above it.

**NBC(AE) Division C, Section 2.3. and NFC(AE) Division C, Section 2.3.**

**Section 2.3. Alternative Solutions**
**2.3.1. Documentation of Alternative Solutions**
(See Note A-2.3.1.)

**2.3.1.1. Documentation**
1) Documentation conforming to this Subsection shall be provided by the person requesting the use of an alternative solution to demonstrate that the proposed alternative solution complies with this Code.

2) The documentation referred to in Sentence (1) shall include
a) a Code analysis outlining the analytical methods and rationales used to determine that a proposed alternative solution will achieve at least the level of performance required by Clause 1.2.1.1.(1)(b) of Division A, and
b) information concerning any special maintenance or operational requirements, including any *building* component commissioning requirements, that are necessary for the alternative solution to achieve compliance with the Code after the *building* is constructed.

3) The Code analysis referred to in Clause (2)(a) shall identify the applicable objectives, functional statements and acceptable solutions, and any assumptions, limiting or restricting factors, testing procedures, engineering studies or *building* performance parameters that will support a Code compliance assessment.

4) The Code analysis referred to in Clause (2)(a) shall include information about the qualifications, experience and background of the person or persons taking responsibility for the design.

5) The information provided under Sentence (3) shall be in sufficient detail to convey the design intent and to support the validity, accuracy, relevance and precision of the Code analysis.

6) Where the design of a *building* includes proposed alternative solutions that involve more than one person taking responsibility for different aspects of the design, the applicant for the *permit* shall identify a single person to co-ordinate the preparation of the design, Code analysis and documentation referred to in this Subsection.

**A-2.3.1. Documentation of Alternative Solutions.** Beyond the purposes of demonstrating compliance and acquiring a building permit, there are other important reasons for requiring that the proponent of an alternative solution submit project documentation (i.e. a compliance report) to the authority having jurisdiction and for the authority having jurisdiction to retain that documentation for a substantial period following the construction of a building:
- Most jurisdictions require that a building be maintained in compliance with the codes under which it was built. Alternative solutions made possible by objective-based codes may have special maintenance requirements, which would be described in the documentation.
- Documentation helps consultants perform code compliance assessments of existing buildings before they are sold and informs current owners or prospective buyers of existing buildings of any limitations pertaining to their future use or development.
- Documentation provides design professionals with the basic information necessary to design changes to an existing building.



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- An alternative solution could be invalidated by a proposed alteration to a building. Designers and regulators must therefore know the details of the particular alternative solutions that were integral to the original design. Complete documentation should provide insight as to why one alternative solution was chosen over another.
- Documentation is the “paper trail” of the alternative solution negotiated between the designer and the regulator and should demonstrate that a rational process led to the acceptance of the alternative solution as an equivalency.
- It is possible that over time a particular alternative solution may be shown to be inadequate. It would be advantageous for a jurisdiction to know which buildings included that alternative solution as part of their design: documentation will facilitate this type of analysis.
- Project documentation provides important information to a forensic team that is called to investigate an accident or why a design failed to provide the level of performance expected.

This subject is discussed in further detail in “Recommended Documentation Requirements for Projects Using Alternative Solutions in the Context of Objective-Based Codes,” which was prepared for the CCBFC Task Group on Implementation of Objective-Based Codes and is available on the NRC's website.

**NBC(AE) Division C, Clause 2.4.1.1.(1)(b)**

**2.4.1.1. General**
1) Except as required by Sentence (4), involvement of a *registered professional* is not required for a *building*
b) classified as a *residential occupancy* that is
i) a single-family dwelling, or
ii) a multiple-family dwelling that contains 4 *dwelling units* or less,

**NFC(AE) Division C, Sentence 2.3.1.2.(1)**

**2.3.1.2. Retention of Documentation**
1) Documentation conforming to Article 2.3.1.1. shall be retained on the premises to which it pertains for each measure that has been approved by the *authority having jurisdiction* and implemented as an alternative solution to acceptable solutions
a) in this Code, or
b) in the NBC(AE) that are attributed to the objectives Fire Safety (OS1), Safety in Use (OS3), Fire Protection of the Building (OP1), or Protection of Adjacent Buildings from Fire (OP3).



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## Application

This interpretation applies to buildings of residential occupancy that require two or more egress stairs as per the acceptable solutions of Division B of the NBC(AE), where the alternative solution path permitted by Sentence 1.2.1.1.(1)(b) of Division A of the NBC(AE), is chosen by the applicant to demonstrate that a single-egress-stair (SES) design complies with the NBC(AE) and the Act.

## Interpretation

1. An AHJ may accept a residential occupancy building of SES design (where the requirements of Division B require two or more exit stairs), provided the following conditions are met:

- The documentation submitted to support the alternative solution is in conformance with Section 2.3. of Division C of the NBC(AE).
- The design provides for an approximately equivalent or greater safety performance required by Division B of the NBC(AE) in the areas defined by the objectives and functional statements attributed to the applicable acceptable solution(s).

2. To ensure ongoing Code compliance following occupancy, the owner must have the issued variance documentation available on the premises for review by the fire SCO to verify the conditions under which the variance was issued have not been contravened.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition and Division C of the National Fire Code – 2023 Alberta Edition

Issued by the Provincial Building and Fire Administrators

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